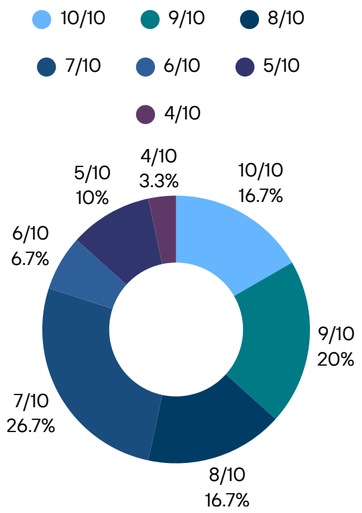
INTEGRATED BUILDING ENVELOPE DESIGN: A BIM BASED WORKFLOW FOR ACHIEVING SUSTAINABILITY

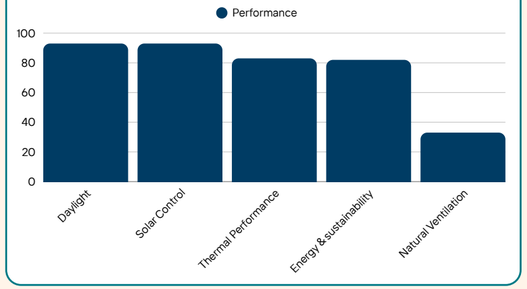
Performance Dashboard

Total score of Case studies

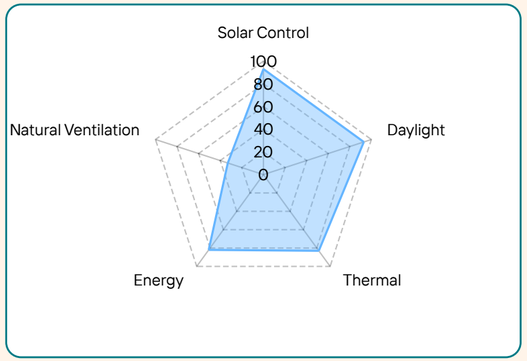


The case studies demonstrate that High-performing projects generally combine multiple strategies including solar shading, high-performance glazing, insulation, secondary skins, daylight optimization and renewable-energy integration.

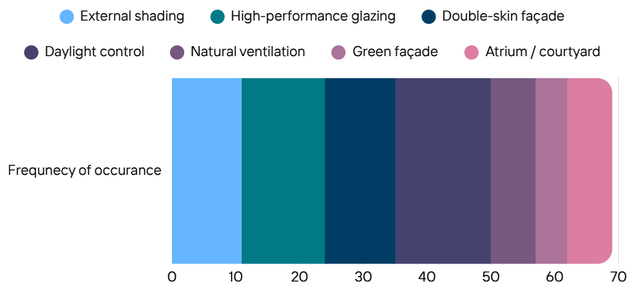
Performance of each Parameter



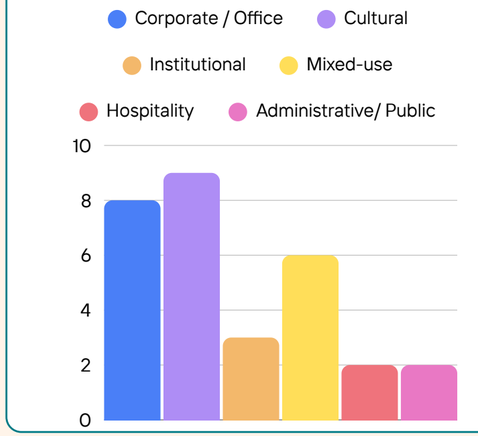
Performance Gap



Frequence of occurance of facade strategies



Performance by Building type



Conclusions & Related Questions

- **Natural ventilation is the weakest-performing parameter across the case studies. This indicates that although contemporary envelopes increasingly optimize solar, daylight and thermal performance, natural ventilation is less consistently incorporated as an envelope-driven strategy.**

1. **Which envelope parameters perform best/worst?**
2. **Which envelope strategies are repeatedly used?**
3. **What strategies should be incorporated ?**
4. **How can BIM transform these proven envelope strategies into an integrated design workflow?**

Rajal Sawant
Sem IX
Goa college of architecture